changing. Thus no e.m.f. will be induced in coil Y to produce a reading.

26 **C** For Loop 3 and loop 4, as the loops did not experience any change in magnetic flux linkage, there were no e.m.f. induced in loop 3 and loop 4. Thus no induced currents flow in loop 3 and loop 4 so the loops did not experience any magnetic force.

Loop 2 experienced a ***decreasing magnetic flux linkage*** through the loop, thus an e.m.f. is induced in the loop. As the loop is closed, an induced current flowed in loop 2. By Lenz's Law, the induced current in loop 2 flow in an anticlockwise direction to produce an effect to oppose the decreasing magnetic flux linkage. Thus by FLHR, there is a downward force on loop 2. (Or By Lenz's Law, there will be a downward force on loop 2 such as to produce an effect to oppose the decreasing magnetic flux linkage)

Loop 1 experienced an ***increasing magnetic flux linkage*** through the loop, thus an e.m.f. is induced in the loop. As the loop is closed, an induced current flowed in loop 1. By Lenz's Law, the induced current in loop 1 flow in a clockwise direction to produce an effect oppose the increasing magnetic flux linkage. Thus by FLHR, there is an upward force on loop 1. (Or By Lenz's Law, there will be a upward force on loop 2 such as to produce an effect to oppose the increasing magnetic flux linkage)

27 **B** When current is forward bias for the single diode, the total resistance is 1 kΩ. The peak current = peak voltage / 1 kΩ